

# Heart Disease Prediction Report

Supervised Machine Learning Classification using Logistic Regression

Machine Learning

Healthcare Analytics

Binary Classification

## PROJECT OVERVIEW

Heart disease remains one of the leading causes of global mortality. Early, accurate detection plays a pivotal role in enabling preventative care and optimizing clinical outcomes. This document details a supervised machine learning methodology designed to evaluate patient physiological parameters and assess the likelihood of heart disease presence using a trained **Logistic Regression** model.

## DATASET OVERVIEW

The classification framework utilizes the standard Heart Disease Dataset, encompassing comprehensive demographic, physical, and clinical diagnostics gathered from patient examinations.

TOTAL INSTANCES

**1,025**

TOTAL FEATURES

**14 Columns**

MISSING VALUES

**0 (Clean)**

## TARGET VARIABLE DISTRIBUTION

The target variable is binary, representing the presence or absence of cardiac conditions. The dataset demonstrates a balanced class distribution, preventing model bias towards any single outcome class.

TARGET: 1 (ISSUE DETECTED)

**526 cases**

TARGET: 0 (HEALTHY)

**499 cases**

TRAIN / TEST SPLIT

**80 / 20 Ratio**

## TECHNICAL STACK

Language: Python

Data Processing: Pandas, NumPy

Model Suite: Scikit-Learn

## FEATURE ATTRIBUTE DEFINITIONS

The 14 clinical features analyzed by the classification pipeline are defined as follows:

| Attribute | Description                                    | Values / Scale                                                 |
|-----------|------------------------------------------------|----------------------------------------------------------------|
| age       | Age of the patient in years                    | Continuous (e.g., 29 to 77)                                    |
| sex       | Biological gender                              | 1 = Male; 0 = Female                                           |
| cp        | Chest pain type experienced                    | Values 0–3 (Typical, Atypical, Non-anginal, Asymptomatic)      |
| trestbps  | Resting blood pressure on admission            | Continuous (mm Hg)                                             |
| chol      | Serum cholestorl level                         | Continuous (mg/dl)                                             |
| fbs       | Fasting blood sugar > 120 mg/dl                | 1 = True; 0 = False                                            |
| restecg   | Resting electrocardiographic results           | Values 0, 1, 2 (Normal, ST-T wave abnormality, LV hypertrophy) |
| thalach   | Maximum heart rate achieved                    | Continuous (bpm)                                               |
| exang     | Exercise induced angina                        | 1 = Yes; 0 = No                                                |
| oldpeak   | ST depression induced by exercise              | Continuous (Relative to rest)                                  |
| slope     | Slope of the peak exercise ST segment          | Values 0–2 (Upsloping, Flat, Downsloping)                      |
| ca        | Number of major vessels colored by fluoroscopy | Values 0–3                                                     |
| thal      | Thalassemia blood disorder test                | 0 = Normal; 1 = Fixed defect; 2 = Reversible defect            |
| target    | Diagnosis outcome label                        | 1 = Heart Disease Detected; 0 = Normal / Healthy               |

## MACHINE LEARNING WORKFLOW

- Data Ingestion & Integrity Check:** Load dataset, perform structural verification, confirm zero null values across all attributes, and evaluate summary statistics.
- Feature Matrix Construction:** Separate independent physiological variables (\$X\$) from the binary diagnosis label (\$Y\$).
- Stratified Data Partitioning:** Perform an 80/20 train-test split utilizing target stratification to ensure both sets retain identical target class distributions.
- Model Fitting & Optimization:** Fit a binary Logistic Regression classifier using optimization algorithms suited for convex loss functions.
- Performance Evaluation:** Compute accuracy scores across both training and unseen testing partitions to measure generalization capacity and prevent overfitting.